TICKET-002 Validation Report

Ticket: TICKET-002 - List Pokémon Endpoint
Endpoint: GET /api/pokemon
Tested At: 2026-04-18T04:53:06Z
Tool: Postman CLI v1.33.6
Status: ✅ PASSED

Summary

- **Total AC**: 6
- **Passed**: 6
- **Failed**: 0

Detailed Results

AC1: Successful Pokémon List Retrieval

Status: ✅ PASSED

Command:

```
postman request GET http://localhost:3000/api/pokemon --response-only
```

Response (Status: 200 OK, Time: 5ms):


```
{
  "success": true,
  "data": [
    {
      "id": 1,
      "name": "bulbasaur",
      "url": "https://pokeapi.co/api/v2/pokemon/1/"
    },
    {
      "id": 2,
      "name": "ivysaur",
      "url": "https://pokeapi.co/api/v2/pokemon/2/"
    }
    // ... 18 more Pokémon (total 20)
  ],
  "pagination": {
    "limit": 20,
    "offset": 0,
    "total": 1350
  }
}
```


Validation:

- ✅ HTTP status code: 200 (expected: 200)
- ✅ Content-Type: application/json; charset=utf-8 (expected: application/json)
- ✅ Response includes "success" field: true
- ✅ Response includes "data" array with Pokémon objects
- ✅ Each Pokémon object has required fields: "id", "name", "url"
- ✅ Response includes "pagination" object with "limit", "offset", and "total" fields
- ✅ Default pagination: limit=20, offset=0, total=1350

AC2: Custom Pagination

Status: ✅ PASSED

Command:

```
postman request GET "http://localhost:3000/api/pokemon?limit=5&offset=10" --
response-only
```

Response:

```
{
  "success": true,
  "data": [
    {"id": 11, "name": "metapod", "url": "https://pokeapi.co/api/v2/pokemon/11/"},
    {"id": 12, "name": "butterfree", "url":
"https://pokeapi.co/api/v2/pokemon/12/"},
    {"id": 13, "name": "weedle", "url": "https://pokeapi.co/api/v2/pokemon/13/"},
    {"id": 14, "name": "kakuna", "url": "https://pokeapi.co/api/v2/pokemon/14/"},
    {"id": 15, "name": "beedrill", "url": "https://pokeapi.co/api/v2/pokemon/15/"}
  ],
  "pagination": {
    "limit": 5,
    "offset": 10,
    "total": 1350
  }
}
```

Validation:

- ✅ Returns exactly 5 Pokémon (expected: 5)
- ✅ Starts from the 11th Pokémon (ID: 11 - metapod, expected offset: 10)
- ✅ Pagination object reflects limit=5 and offset=10
- ✅ Data correctly starts at offset position (ID 11 is the 11th Pokémon)

AC3: Limit Validation

Status: ✅ PASSED

Command:

```
postman request GET "http://localhost:3000/api/pokemon?limit=150" --response-only
```

Response Summary:

```
{
  "success": true,
  "data": [ /* 100 Pokémon objects (IDs 1-100) */ ],
  "pagination": {
    "limit": 100,
    "offset": 0,
    "total": 1350
  }
}
```

Validation:

- ✅ Requested limit: 150, capped at: 100 (max allowed)
- ✅ Returns exactly 100 Pokémon (count verified)
- ✅ Pagination object shows limit: 100
- ✅ Last Pokémon in response: ID 100 (voltorb)

AC4: Invalid Pagination Parameters

Status: ✅ PASSED

Command:

```
postman request GET "http://localhost:3000/api/pokemon?limit=-5" --response-only
```

Response (Status: 400 Bad Request, Time: 5ms):

```
{
  "success": false,
  "error": {
    "code": "INVALID_PARAMETERS",
    "message": "Limit and offset must be non-negative integers"
  }
}
```

Validation:

- ✅ HTTP status code: 400 (expected: 400)
- ✅ Response includes "success" field: false
- ✅ Response includes "error" object with "code" and "message"
- ✅ Error code: INVALID_PARAMETERS
- ✅ Error message indicates invalid parameters: "Limit and offset must be non-negative integers"

AC5: Data Source

Status: ✅ PASSED

Command:

```
postman request GET "http://localhost:3000/api/pokemon?limit=3" --response-only
```

Response:

```
{
  "success": true,
  "data": [
    {"id": 1, "name": "bulbasaur", "url": "https://pokeapi.co/api/v2/pokemon/1/"},
    {"id": 2, "name": "ivysaur", "url": "https://pokeapi.co/api/v2/pokemon/2/"},
    {"id": 3, "name": "venusaur", "url": "https://pokeapi.co/api/v2/pokemon/3/"}
  ],
  "pagination": {"limit": 3, "offset": 0, "total": 1350}
}
```

Validation against PokéAPI:

```
curl -s "https://pokeapi.co/api/v2/pokemon?limit=3&offset=0"
```

PokéAPI Response:

```
{
  "count": 1350,
  "results": [
    {"name": "bulbasaur", "url": "https://pokeapi.co/api/v2/pokemon/1/"},
    {"name": "ivysaur", "url": "https://pokeapi.co/api/v2/pokemon/2/"},
    {"name": "venusaur", "url": "https://pokeapi.co/api/v2/pokemon/3/"}
  ]
}
```

Validation:

- ✅ Pokémon names match PokéAPI data: bulbasaur, ivysaur, venusaur
- ✅ URLs match PokéAPI format: https://pokeapi.co/api/v2/pokemon/{id}/
- ✅ Total count matches PokéAPI: 1350
- ✅ Data is correctly sourced from PokéAPI (https://pokeapi.co/api/v2/pokemon)

AC6: Response Time

Status: ✅ PASSED

Command:

```
postman request GET "http://localhost:3000/api/pokemon?limit=50&offset=100"
```

Response Time: 657ms

Validation:

- ✅ Response time: 657ms (expected: < 2000ms)
- ✅ Performance requirement met (well under the 2-second threshold)

Additional Test Coverage

Edge Case Testing

All the following scenarios were implicitly tested:

- **Default pagination**: Works correctly with limit=20, offset=0
- **Large dataset handling**: Successfully handles requests with max limit (100)
- **Offset positioning**: Correctly skips records based on offset parameter
- **Error handling**: Proper 400 error for negative parameters

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Recommendations

All acceptance criteria have been successfully validated. The endpoint is working as expected:

1. ✅ **Response Structure**: All required fields present and correctly formatted
2. ✅ **Pagination**: Default, custom, and max limit scenarios work correctly
3. ✅ **Validation**: Invalid parameters are properly rejected with appropriate error messages
4. ✅ **Data Integrity**: Data matches PokéAPI source
5. ✅ **Performance**: Response times are well within acceptable limits
6. ✅ **Error Handling**: Proper HTTP status codes and error messages

Notes:

- Response times for cached requests are significantly faster (5ms) vs first call (657ms)
- The endpoint properly caps the limit at 100 as per specification
- Error messages are clear and informative for invalid inputs
- The API successfully integrates with external PokéAPI service

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Test Completed: 2026-04-18T04:53:06Z
Tester: Postman CLI Automation
Overall Result: ✅ ALL ACCEPTANCE CRITERIA PASSED